

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Hybrid Ensemble Machine Learning with Explainable AI for Early Prediction of Parkinson's Disease Severity

Archit Sanoria

2210991334

Supervised By

Ajay Katiyar

Department of Computer Science and Engineering,
Chitkara University, Punjab

- **What is Parkinson's Disease?**
 - o Progressive neurological disorder affecting 10+ million people worldwide
 - o Causes tremor, rigidity, bradykinesia, speech impairment
 - o Severity measured using UPDRS (Unified Parkinson's Disease Rating Scale)
- **The Problem with Clinical Monitoring:**
 - o Frequent hospital visits are expensive and inconvenient
 - o Daily symptom variations are never captured
 - o Voice biomarkers offer a remote, non-invasive alternative
- **Key Statistics:**
 - o 10 Million+ patients worldwide
 - o 5,875 voice recordings from 42 subjects (Oxford dataset)
 - o UPDRS Severity Score range: 7–55 scale
- **Key Innovation:**
 - o Subject-independent evaluation: No patient data leakage between training and testing sets

Literature Review & Problem Statement

- **Critical Flaw in Existing Research:**
 - o Most studies split 5,875 recordings randomly (record-level split)
 - o Same patient's recordings appear in BOTH training & test sets
 - o This is DATA LEAKAGE — models memorize patient identity
 - o Reported R^2 values as high as 0.91 are inflated artifacts

Study	Method	R^2	Leakage?
Tsanas et al. (2010)	Sparse Bayesian Regression	0.76	Yes
Grover et al. (2018)	RF + Gradient Boosting	0.91	Yes
Ali et al. (2019)	XGBoost 10-fold CV	0.87	Yes
Sakar et al. (2019)	SVM + Wavelets (LOSO)	0.79	No
Proposed (XGBoost)	Subject-wise Hold-out	0.8767	No

- **Dataset: Oxford Parkinson's Disease Telemonitoring**
 - o 42 Patients — Parkinson's diagnosed
 - o 5,875 Records — ~140 per patient
 - o 20 Features — Age, sex, 16 acoustic + motor
 - o 6 Months — Home monitoring period
- **Methodology Pipeline:**
 - o 1. Load & Preprocess Data (Drop test_time; retain motor_UPDRS)
 - o 2. Patient-Level 80/20 Split (33 patients train | 9 patients test)
 - o 3. Train Base Learners (RF, XGBoost, SVR, MLP)
 - o 4. GroupKFold OOF Predictions (Subject-wise, 5-fold)
 - o 5. Train Meta-Learner XGBoost on OOF meta-feature matrix
 - o 6. Evaluate & SHAP Analysis (R^2 , MAE, RMSE, Wilcoxon)

- **Four Base Learners:**
 - o Random Forest: `n_estimators=200`, Bootstrap sampling, Unscaled features
 - o XGBoost: `n_estimators=300`, `lr=0.05`, `max_depth=5`, L1/L2 regularization
 - o SVR: RBF kernel `C=1.0`, StandardScaler pipeline fitted on train only
 - o MLP: Layers (128, 64), ReLU + Adam optimizer, Max 1000 iterations
- **Hybrid Stacking Ensemble Architecture:**
 - o Training Data: 4,640 recordings from 33 patients
 - o Level 1 — Base Learners generate GroupKFold OOF Predictions (patient-disjoint)
 - o Meta-Feature Matrix: $4,640 \times 4$ (OOF predictions from all base learners)
 - o Level 2 — Meta-Learner (XGBoost) trained on OOF matrix
 - o Final predictions on 1,235 test recordings from 9 unseen patients
 - o Key advantage: Subject-wise GroupKFold ensures NO patient identity leakage at ANY stage of the stacking pipeline

Results – Individual Model Performance

Model	CV R^2 (Mean $\pm$ Std)	Test R^2	Test MAE	Test RMSE
XGBoost	0.739 ± 0.127	0.8767	2.34	3.20
SVR	0.731 ± 0.077	0.7850	3.23	4.22
Random Forest	0.728 ± 0.135	0.7798	3.13	4.27
MLP	0.713 ± 0.076	0.6024	4.42	5.74

- **Key Finding:**
 - o XGBoost achieves $R^2 = 0.8767$ on 9 completely unseen patients
 - o Average error: only 2.34 UPDRS points — clinically acceptable
 - o All differences statistically significant ($p < 0.0001$)
 - o Stacking Ensemble $R^2 = 0.7346$ — XGBoost alone outperforms the ensemble

Stacking Ensemble & Statistical Significance

- **All Models — R^2 Comparison:**
 - o XGBoost → Test $R^2 = 0.8767$ (Best individual model)
 - o SVR → Test $R^2 = 0.7850$
 - o Random Forest → Test $R^2 = 0.7798$
 - o Stacking → Test $R^2 = 0.7346$ (Ensemble underperforms)
 - o MLP → Test $R^2 = 0.6024$

Comparison	p-value
XGBoost vs RF	<0.0001
XGBoost vs SVR	<0.0001
XGBoost vs MLP	<0.0001
XGBoost vs Stacking	<0.0001
Stacking vs RF	0.0295
Stacking vs SVR	0.0224

- **Why did stacking underperform XGBoost alone?**
 - o Only 33 subjects available for meta-learning
 - o All base learners fail on the same patient groups
 - o XGBoost dominance makes weaker models add noise
 - o Leakage-free stacking reveals true ensemble limits

- **Top 10 Features by Mean Absolute SHAP Value:**
 - o motor_UPDRS (8.29) — Co-measured motor sub-score dominates predictions
 - o age (1.18) — Older patients consistently show higher severity
 - o sex (0.45) — Sex-linked variation in Parkinson's progression
 - o DFA (0.38) — Vocal fractal irregularity — key nonlinear biomarker
 - o Jitter(Abs) (0.23) — Absolute pitch perturbation — vocal tremor indicator
 - o Shimmer:APQ11 (0.08) — Amplitude perturbation — amplitude instability
 - o PPE (0.07) — Pitch Period Entropy — voice complexity measure
 - o Jitter:RAP (0.06) — Relative Average Perturbation in fundamental frequency
 - o HNR (0.06) — Harmonics-to-Noise Ratio — overall voice quality
 - o RPDE (0.05) — Recurrence Period Density Entropy — nonlinear vocal dynamics
- **Waterfall Example: Actual UPDRS=42.1 → Predicted=41.7**
 - o Baseline 24.4 + motor_UPDRS (+14.2) + age (+3.1) + DFA (+0.8) – HNR (–0.6) = 41.7

- **Conclusion:**
 - o XGBoost achieves $R^2 = 0.8767$ on 9 completely unseen patients with MAE = 2.34 UPDRS points
 - o Subject-independent evaluation exposes data leakage in all prior work (record-level R^2 collapses to -1.64 to -0.22 on truly unseen patients)
 - o Stacking ensemble ($R^2 = 0.7346$) underperforms XGBoost alone when cohort size is constrained
 - o SHAP analysis enables clinical interpretability: motor_UPDRS, age, DFA, and PPE are the dominant predictors
 - o All performance differences statistically significant (Wilcoxon, $p < 0.05$)
- **Future Scope:**
 - o External validation on different country/demographic datasets
 - o Leave-One-Subject-Out (LOSO) validation for stricter evaluation
 - o Bayesian hyperparameter optimization for XGBoost
 - o LSTM deep learning for longitudinal multi-recording analysis
 - o Mobile app integration for real-time home-based severity assessment

Thank You